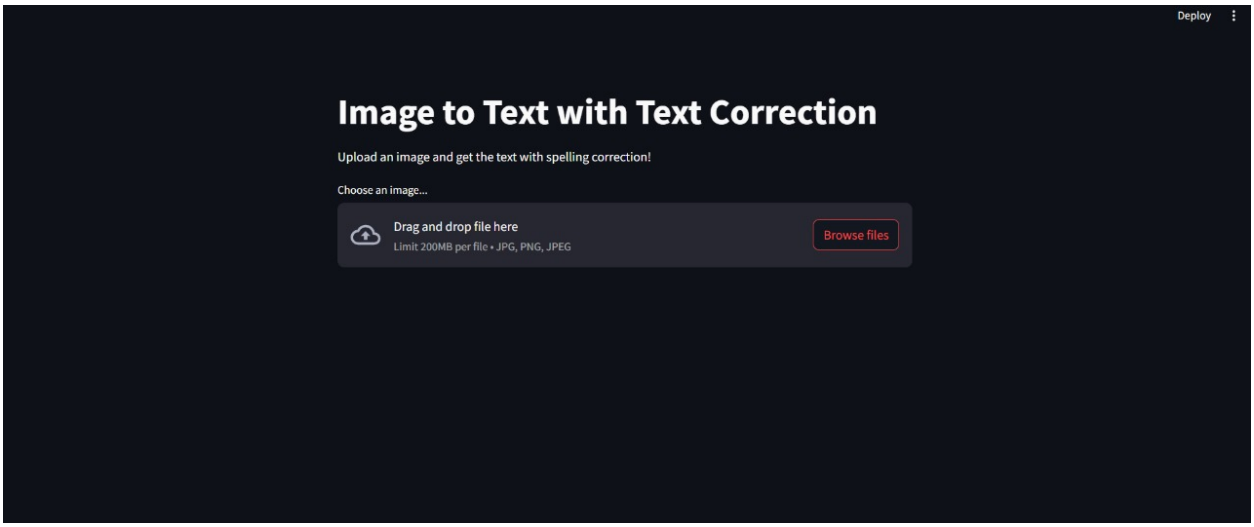


Report

Image-to-Text Conversion and Text Correction System Report

Introduction

In this report, we present a project aimed at extracting text from images using EasyOCR and performing spell checking using the PySpellChecker library. Additionally, we have developed a web application using Streamlit to provide a user-friendly interface for utilizing these functionalities.

Project Overview

Our project encompasses the following main components:

1. Text extraction from images using EasyOCR.
2. Spell checking using PySpellChecker.
3. Development of a web application using Streamlit.

Methodology

Text Extraction with EasyOCR

EasyOCR is a Python library that allows for easy and efficient text extraction from images. We utilize its capabilities to extract text from images uploaded by the user through the web application.

Spell Checking with PySpellChecker

PySpellChecker is a Python library that provides efficient spell checking functionality. After extracting text from images, we pass it through PySpellChecker to identify and correct any spelling errors present in the text.

Web Application Development with Streamlit

Streamlit is a popular framework for building web applications with Python. We leverage Streamlit to create a user-friendly interface where users can upload images, extract text, perform spell checking, and view the corrected text.

Implementation Details

Required Libraries:

```
#!pip3 install torch torchvision torchaudio --index-url https://
!pip install easyocr
!pip install --user opencv-python
pip install pyspellchecker
```

Import Libraries:

```
import easyocr
import cv2
from matplotlib import pyplot as plt
import pandas as pd
from spellchecker import SpellChecker
```

Text Extraction

```
# Provide your image path to image_path variable
image_path= r"C:\Users\Dell\OneDrive\Pictures\Screenshots\Scr
eenshot 2024-04-26 030414.png"

reader= easyocr.Reader(['en'],gpu=False)
result=reader.readtext(image_path)
print(result)
#result contain String and co-ordinate of the text
```

Covert it to Data Frame to better understand :

```
#Covert the ouput into df to better understand the text
df = pd.DataFrame(result, columns= ['Cor','text','confidence'])
s = [str(x) for x in df["text"] ]
s1 = " ".join(s)
#s1 contain the Text which we extracted from the image
```

Spell Checking

```
spell = SpellChecker()
l = ""
for word in s1.split() :
    l = l + " " + spell.correction(word)
print(corrected_text)
```

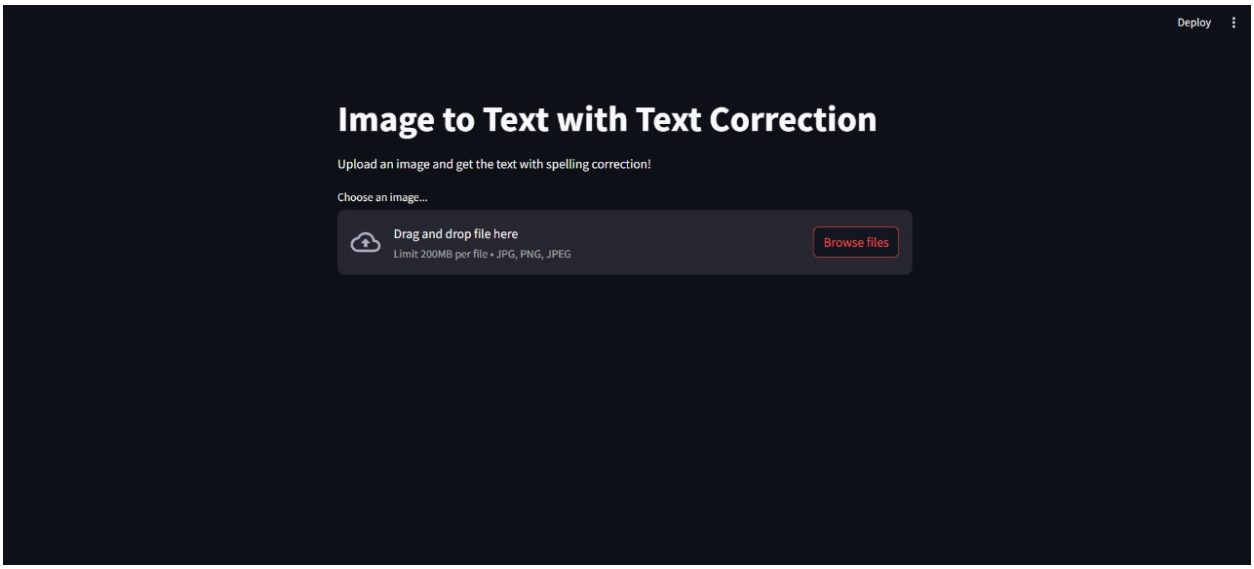
Streamlit Web Application

```
pythonCopy code
# Code snippet for Streamlit web application
import streamlit as st

def main():
    # Title and description
    st.title("Image to Text with Text Correction")
    st.write("Upload an image and get the text with spelling
correction!")

    # Define directory to store uploaded images
    upload_dir = "uploaded_images"
    os.makedirs(upload_dir, exist_ok=True)

    # File uploader
    uploaded_file = st.file_uploader("Choose an image...", ty
pe=["jpg", "png"])

    if uploaded_file is not None:
        # Store the uploaded file in the specified directory
        image_path = os.path.join(upload_dir, uploaded_file.n
ame)
        with open(image_path, "wb") as f:
            f.write(uploaded_file.getvalue())

        # Display uploaded image
        image = Image.open(uploaded_file)
        st.image(image, caption="Uploaded Image", use_column_
width=True)

        # Store full path of the uploaded image
                 full_image_path = os.path.abspath(image_pat
h)

    #Passing full image to our easyocr module
```

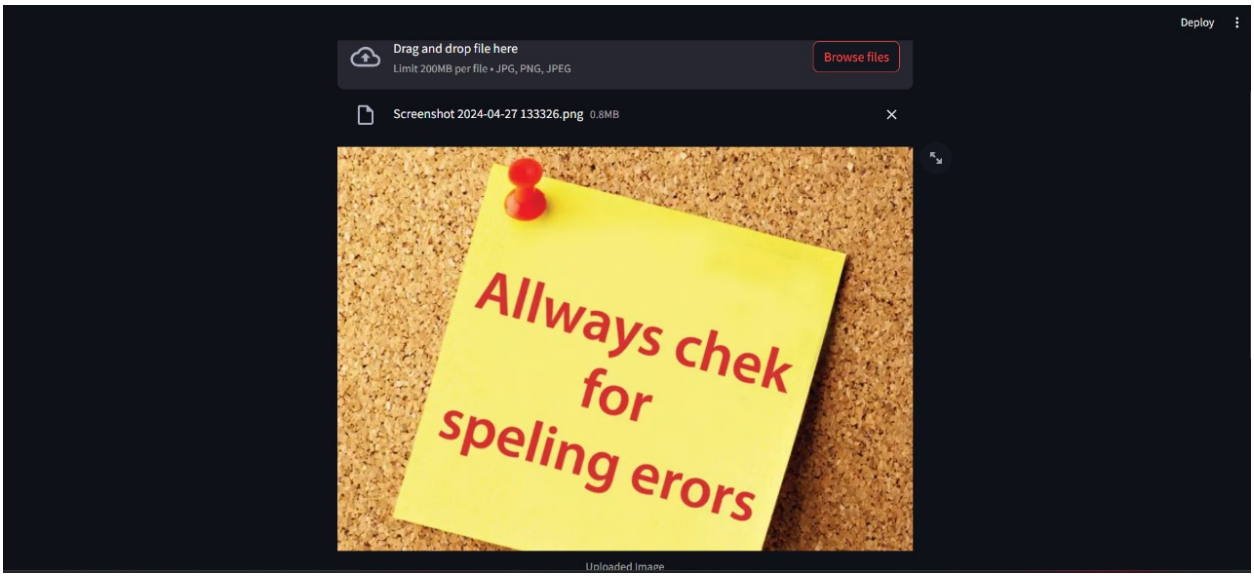
```
        reader = easyocr.Reader(['en'], gpu=False)
        result = reader.readtext(full_image_path)

    # coverting our text into data frame to better understand
        df = pd.DataFrame(result, columns=['Cor', 'text', 'co
nfidence'])
        s = [str(x) for x in df["text"]]
        s1 = " ".join(s)
        st.title("Extracted Text from the image: ")
        st.code(s1)


    # spellijg checker
        spell = SpellChecker()
        l = ""
        for word in s1.split():
            l = l + " " + spell.correction(word)
        st.title("Spelling Corrected text after Extraction:
")
        st.code(l)

if _name_ == "_main_":
    main()
```

Results

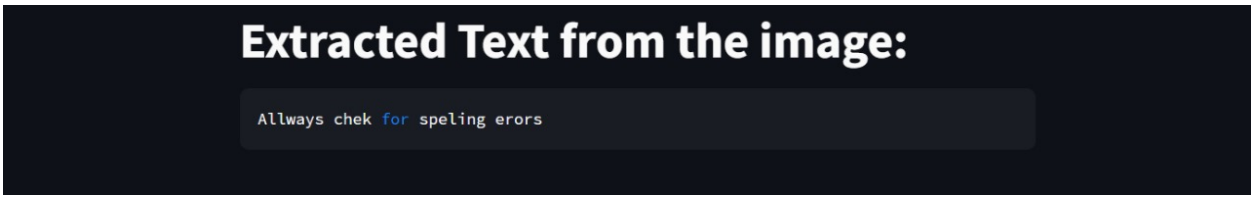
Web Application Interface



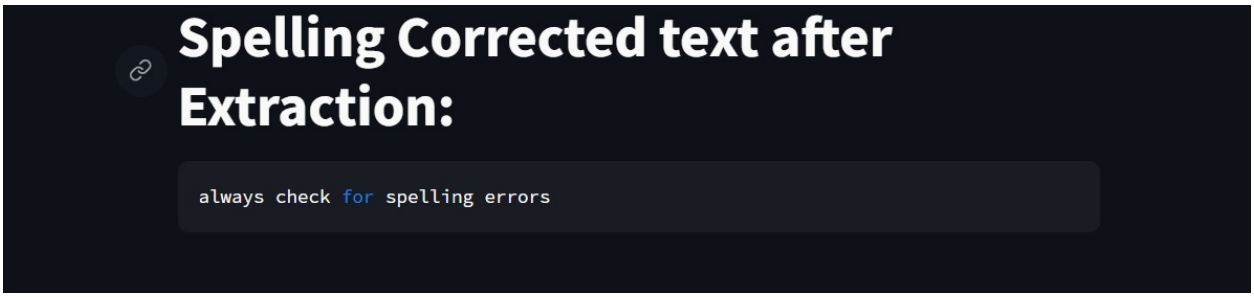
Uploading Image



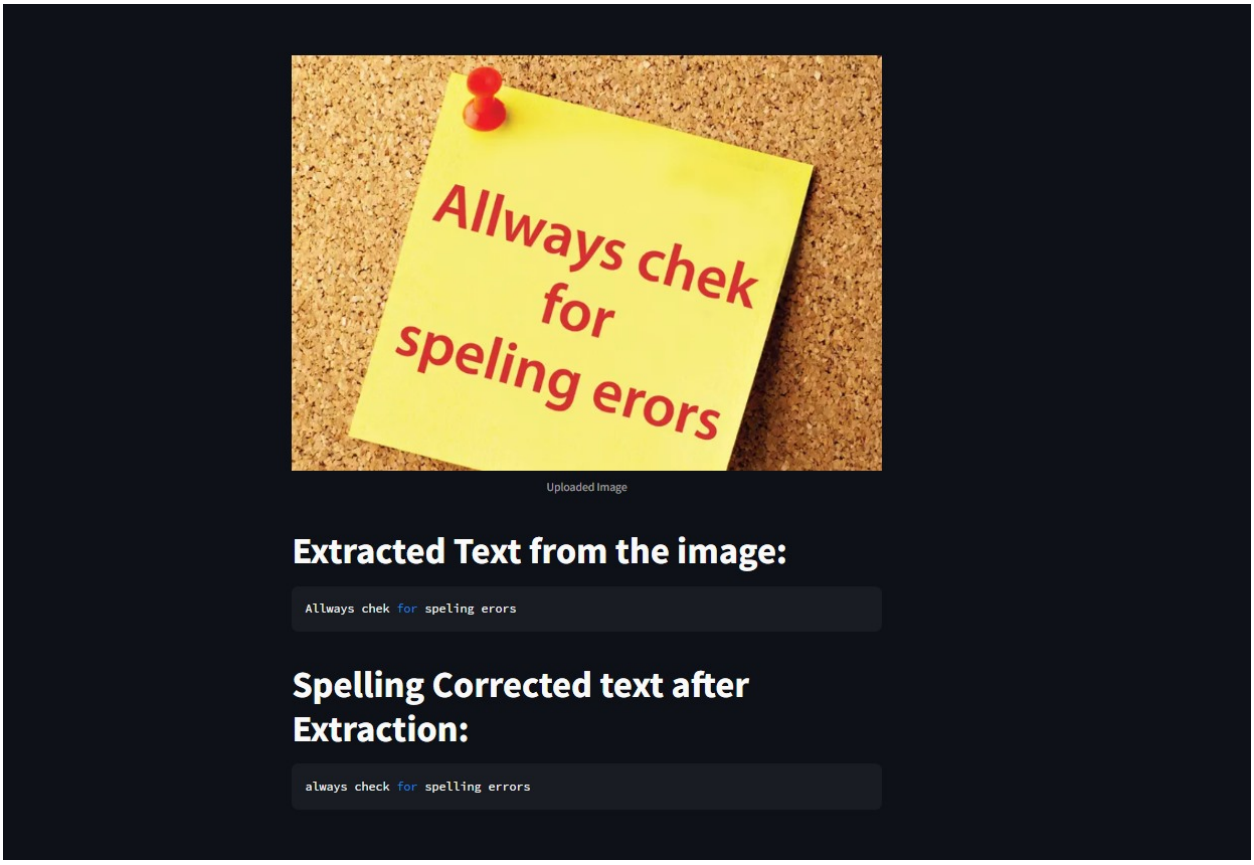
Text Extraction Output



Spell Checking Output



Web Application Screenshots



Conclusion

Our project successfully demonstrates the integration of EasyOCR for text extraction, PySpellChecker for spell checking, and Streamlit for web application development. The combination of these technologies provides a powerful tool for extracting and correcting text from images in a user-friendly manner.